

Course 5043C

Semester V

Theory

4 Credits

Telecommunication Networks

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043C as Telecommunication Networks in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Data Communication course and polytechnic networking curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Network designs, IP planning, protocol configurations and security policies must be based on approved engineering plans, certified hardware data, current industry standards and competent professional review.

Verified source note: Verified sources support OSI/TCP-IP models, switching, transmission media, and routing. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Telecommunication Networks studies the principles, architectures and protocols used to exchange data across local and wide area networks. It develops the ability to analyze the OSI and TCP/IP reference models, evaluate switching techniques and transmission media, understand IP addressing and routing, and coordinate networked solutions while respecting the technical and security boundaries of electronic and communication engineering.

Malayalam support: □ handbook □□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□□.

Prerequisite foundation

Basic electronics, analog communication, digital electronics and computer fundamentals.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of data communication, network topologies and reference models.
2. Analyze the characteristics of transmission media and error detection/correction techniques.
3. Evaluate switching techniques, flow control and medium access control (MAC) protocols.
4. Explain the systematic operation of the network and transport layers including IP and TCP.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO നോട്ടീസ് exam-ന് മുമ്പായി നോട്ടീസ് നോട്ടീസ് നോട്ടീസ് നോട്ടീസ്. Define, explain, apply നോട്ടീസ് action words നോട്ടീസ്.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals of data communication and network reference models.	Understand
CO2	Analyze transmission media, error control and data link layer protocols.	Apply
CO3	Evaluate switching techniques and medium access control for networks.	Apply
CO4	Explain the operation of network and transport layer protocols like IP and TCP.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Data Communication and Reference Models	Data communication components; Network topologies: Star, Mesh, Ring, Bus; Network categories: LAN, MAN, WAN; OSI Reference Model; TCP/IP Model; Comparison of models.	Approx. 14 hours	CO1	Module 1
2	Physical and Data Link Layers	Transmission media: Guided (Twisted pair, Coaxial, Fiber) and Unguided; Data Link Layer: Framing, Error detection (CRC, Parity), Flow control (Stop-and-wait, Sliding window); HDLC basics.	Approx. 16 hours	CO2	Module 2
3	Switching and Medium Access	Switching techniques: Circuit switching, Packet switching (Datagram and Virtual circuit); Medium Access Control (MAC): ALOHA, CSMA/CD, CSMA/CA; Ethernet standards; Wireless LAN (802.11).	Approx. 16 hours	CO3	Module 3
4	Network and Transport Layers	Network Layer: IP Addressing (IPv4 vs IPv6), Subnetting; Routing algorithms: Distance Vector, Link State; Transport Layer: TCP and UDP features; Congestion control; Application layer overview (HTTP, DNS).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Data Communication and Reference Models

Data communication components; Network topologies: Star, Mesh, Ring, Bus; Network categories: LAN, MAN, WAN; OSI Reference Model; TCP/IP Model; Comparison of models.

CO1 · Approx. 14 hours

Physical and Data Link Layers

Transmission media: Guided (Twisted pair, Coaxial, Fiber) and Unguided; Data Link Layer: Framing, Error detection (CRC, Parity), Flow control (Stop-and-wait, Sliding window); HDLC basics.

CO2 · Approx. 16 hours

Switching and Medium Access

Switching techniques: Circuit switching, Packet switching (Datagram and Virtual circuit); Medium Access Control (MAC): ALOHA, CSMA/CD, CSMA/CA; Ethernet standards; Wireless LAN (802.11).

CO3 · Approx. 16 hours

Network and Transport Layers

Network Layer: IP Addressing (IPv4 vs IPv6), Subnetting; Routing algorithms: Distance Vector, Link State; Transport Layer: TCP and UDP features; Congestion control; Application layer overview (HTTP, DNS).

CO4 · Approx. 14 hours

MODULE 1

Data Communication and Reference Models

Data communication components; Network topologies: Star, Mesh, Ring, Bus; Network categories: LAN, MAN, WAN; OSI Reference Model; TCP/IP Model; Comparison of models.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

Identify core terms, explain the principle and complete the stated application or practical

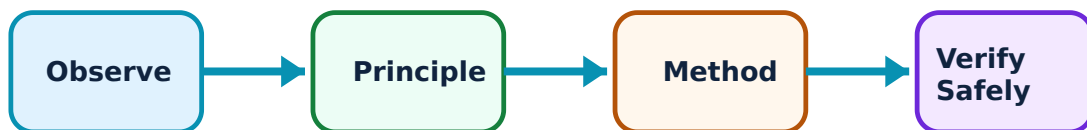
outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Data Communication and Reference Models: identify the system, state its principle, follow the correct method and verify the result safely.

1. Network Fundamentals–

Data communication involves the exchange of data between two nodes. A network is a collection of devices connected by communication links. Topologies define the physical or logical arrangement of these links. LANs cover small areas, while WANs span large distances.

Study and application method

Sketch the 'Star' and 'Mesh' topologies; explain the difference between 'LAN' and 'WAN' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'Star' and 'Mesh' topologies; explain the difference between 'LAN' and 'WAN' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഡയഗ്രാം / സർക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ വ്യക്തമായി വിശദീകരിക്കുക. ഫോർമുലാ റിസൾട്ട് അപ്ലിക്കേഷൻ വ്യക്തമായി വിശദീകരിക്കുക. Technical terms English-ൽ വ്യക്തമായി വിശദീകരിക്കുക.

Common mistake / precaution: Improper topology selection can lead to single points of failure. Do not design networks without authorised redundancy and scalability analysis.

2. Reference Models (OSI and TCP/IP)–

Reference models provide a framework for network communication. The OSI model has 7 layers (Physical to Application), while the TCP/IP model has 4 or 5 layers. Each layer has specific functions and protocols to ensure interoperability.

Study and application method

List the 7 layers of the 'OSI Model' and state one function for each; compare 'OSI' and 'TCP/IP' models in a conceptual table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the 7 layers of the 'OSI Model' and state one function for each; compare 'OSI' and 'TCP/IP' models in a conceptual table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോശ്ഠ concept കോശ്ഠം കോശ്ഠം കോശ്ഠം. കോശ്ഠ diagram / circuit / formula / procedure കോശ്ഠം കോശ്ഠം. കോശ്ഠ result കോശ്ഠം application കോശ്ഠം കോശ്ഠം കോശ്ഠം കോശ്ഠം. Technical terms English-ഈ കോശ്ഠം കോശ്ഠം കോശ്ഠം.

Common mistake / precaution: Layered models are abstractions. Do not assume all protocols map perfectly to a single layer without authorised review of the protocol specification.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Physical and Data Link Layers

Transmission media: Guided (Twisted pair, Coaxial, Fiber) and Unguided; Data Link Layer: Framing, Error detection (CRC, Parity), Flow control (Stop-and-wait, Sliding window); HDLC basics.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

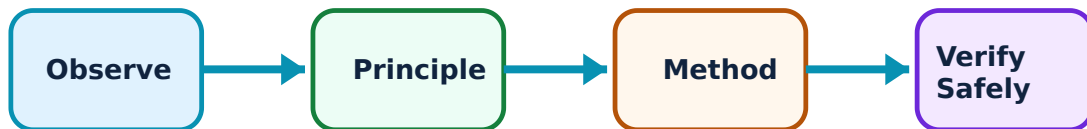
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Physical and Data Link Layers: identify the system, state its principle, follow the correct method and verify the result safely.

1. Transmission Media–

Transmission media carry the signal from sender to receiver. Guided media use physical cables, with Fiber Optics offering the highest bandwidth. Unguided media use wireless signals (Radio, Microwave, Infrared).

Study and application method

Compare 'Twisted Pair' and 'Fiber Optic' cables conceptually; explain the advantage of 'Wireless' communication for mobile users.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Twisted Pair' and 'Fiber Optic' cables conceptually; explain the advantage of 'Wireless' communication for mobile users.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: concept diagram / circuit / formula / procedure result application. Technical terms English- Malayalam.

Common mistake / precaution: Signal attenuation and interference limit transmission distance. Do not deploy media without authorised link budget analysis and shielding requirements.

2. Data Link Control–

The Data Link Layer ensures reliable point-to-point communication. It uses 'Framing' to divide data into packets. Error detection like 'CRC' (Cyclic Redundancy Check) identifies corrupted data. Flow control mechanisms prevent a fast sender from overwhelming a slow receiver.

Study and application method

Explain the 'CRC' error detection process conceptually; describe the 'Sliding Window' flow control mechanism.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'CRC' error detection process conceptually; describe the 'Sliding Window' flow control mechanism.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High error rates can degrade network throughput. All link-layer designs must follow authorised error-correction and retransmission strategies.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Switching and Medium Access

Switching techniques: Circuit switching, Packet switching (Datagram and Virtual circuit); Medium Access Control (MAC): ALOHA, CSMA/CD, CSMA/CA; Ethernet standards; Wireless LAN (802.11).

Approx. 16 hours

CO3

Understand · Apply

Learning goals

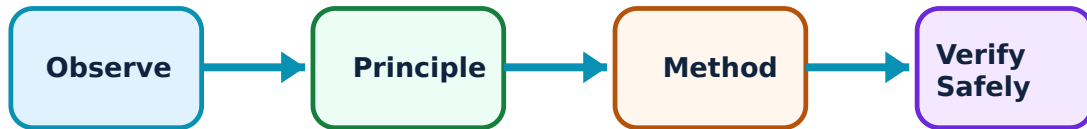
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Switching and Medium Access: identify the system, state its principle, follow the correct method and verify the result safely.

1. Switching Techniques–

Switching directs data through a network. Circuit switching establishes a dedicated path (e.g., PSTN). Packet switching divides data into packets that share links, improving efficiency. Datagram switching is connectionless, while Virtual Circuits are connection-oriented.

Study and application method

Compare 'Circuit Switching' and 'Packet Switching' in a conceptual table; explain the 'Datagram' approach to routing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Circuit Switching' and 'Packet Switching' in a conceptual table; explain the 'Datagram' approach to routing.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ഈ diagram / circuit / formula / procedure ഈ result application ഈ technical terms English-ൽ എങ്ങനെ വിശദീകരിക്കണം. Technical terms English-ൽ എങ്ങനെ വിശദീകരിക്കണം.

Common mistake / precaution: Packet loss and jitter can affect real-time applications. Do not use packet-switched networks for time-critical control without authorised Quality of Service (QoS) mechanisms.

2. Medium Access Control (MAC)–

MAC protocols coordinate access to a shared medium. CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is used in wired Ethernet. CSMA/CA (Collision Avoidance) is used in wireless networks (Wi-Fi) to handle the 'Hidden Node' problem.

Study and application method

Describe the 'CSMA/CD' process conceptually; explain why 'CSMA/CA' is used in wireless LANs instead of collision detection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'CSMA/CD' process conceptually; explain why 'CSMA/CA' is used in wireless LANs instead of collision detection.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കണ്ടെത്തുക concept കണ്ടെത്തുക കണ്ടെത്തുക. കണ്ടെത്തുക diagram / circuit / formula / procedure കണ്ടെത്തുക കണ്ടെത്തുക. കണ്ടെത്തുക result കണ്ടെത്തുക application കണ്ടെത്തുക കണ്ടെത്തുക കണ്ടെത്തുക. Technical terms English-ൽ കണ്ടെത്തുക കണ്ടെത്തുക.

Common mistake / precaution: Collisions reduce network efficiency. All shared-medium designs must follow authorised collision-domain limits and access-control rules.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Network and Transport Layers

Network Layer: IP Addressing (IPv4 vs IPv6), Subnetting; Routing algorithms: Distance Vector, Link State; Transport Layer: TCP and UDP features; Congestion control; Application layer overview (HTTP, DNS).

Approx. 14 hours

CO4

Understand · Apply

Learning goals

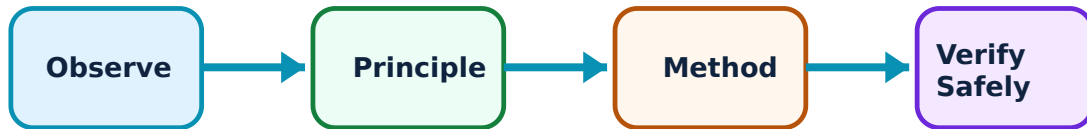
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Network and Transport Layers: identify the system, state its principle, follow the correct method and verify the result safely.

1. IP Addressing and Routing–

The Network Layer handles end-to-end delivery. IP addresses identify devices. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses to handle the growing number of devices. Routing algorithms determine the best path for data.

Study and application method

Differentiate between 'IPv4' and 'IPv6' conceptually; explain the purpose of 'Subnetting' in IP networks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Differentiate between 'IPv4' and 'IPv6' conceptually; explain the purpose of 'Subnetting' in IP networks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. ഡയഗ്രാം / സിരക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ ഉപയോഗിക്കുക. ഫലം വ്യക്തമായി പറയുക. result application ഉപയോഗിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Common mistake / precaution: Incorrect IP configuration can cause network isolation or conflicts. All IP planning must follow authorised address-allocation and subnetting schemes.

2. Transport and Application Protocols–

The Transport Layer provides process-to-process delivery. TCP is connection-oriented and reliable, while UDP is connectionless and fast. Congestion control prevents network collapse. Application layer protocols like HTTP (web) and DNS (naming) provide user services.

Study and application method

Compare 'TCP' and 'UDP' in a conceptual table; describe the role of 'DNS' in the internet.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'TCP' and 'UDP' in a conceptual table; describe the role of 'DNS' in the internet.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result application ന്നുള്ള technical terms English-ൽ എഴുതുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Network congestion can lead to significant delays. Do not deploy high-traffic applications without authorised congestion-avoidance and load-balancing strategies.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Data Communication and Reference Models

Use the concept flow to revise observation, principle, method and verification.

Module 2: Physical and Data Link Layers

Use the concept flow to revise observation, principle, method and verification.

Module 3: Switching and Medium Access

Use the concept flow to revise observation, principle, method and verification.

Module 4: Network and Transport Layers

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare OSI and TCP/IP models in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a 'Star' network topology showing a central switch and four nodes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the frame format of a 'Standard Ethernet' frame conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the 'Subnet Mask' conceptually for a given IP address and number of subnets.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a network device in your environment (e.g., Router, Switch) and note its function.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Data Communication and Reference Models

Explain Network Fundamentals and state one correct method, application or precaution. –

Data communication involves the exchange of data between two nodes. A network is a collection of devices connected by communication links. Topologies define the physical or logical arrangement of these links. LANs cover small areas, while WANs span large distances.

Answer framework: Sketch the 'Star' and 'Mesh' topologies; explain the difference between 'LAN' and 'WAN' conceptually.

Important point: Improper topology selection can lead to single points of failure. Do not design networks without authorised redundancy and scalability analysis.

Explain Reference Models (OSI and TCP/IP) and state one correct method, application or precaution. –

Reference models provide a framework for network communication. The OSI model has 7 layers (Physical to Application), while the TCP/IP model has 4 or 5 layers. Each layer has specific functions and protocols to ensure interoperability.

Answer framework: List the 7 layers of the 'OSI Model' and state one function for each; compare 'OSI' and 'TCP/IP' models in a conceptual table.

Important point: Layered models are abstractions. Do not assume all protocols map perfectly to a single layer without authorised review of the protocol specification.

Module 2 — Physical and Data Link Layers

Explain Transmission Media and state one correct method, application or precaution. –

Transmission media carry the signal from sender to receiver. Guided media use physical cables, with Fiber Optics offering the highest bandwidth. Unguided media use wireless signals (Radio, Microwave, Infrared).

Answer framework: Compare 'Twisted Pair' and 'Fiber Optic' cables conceptually; explain the advantage of 'Wireless' communication for mobile users.

Important point: Signal attenuation and interference limit transmission distance. Do not deploy media without authorised link budget analysis and shielding requirements.

Explain Data Link Control and state one correct method, application or precaution. –

The Data Link Layer ensures reliable point-to-point communication. It uses 'Framing' to divide data into packets. Error detection like 'CRC' (Cyclic Redundancy Check) identifies corrupted data. Flow control mechanisms prevent a fast sender from overwhelming a slow receiver.

Answer framework: Explain the 'CRC' error detection process conceptually; describe the 'Sliding Window' flow control mechanism.

Important point: High error rates can degrade network throughput. All link-layer designs must follow authorised error-correction and retransmission strategies.

Module 3 — Switching and Medium Access

Explain Switching Techniques and state one correct method, application or precaution. —

Switching directs data through a network. Circuit switching establishes a dedicated path (e.g., PSTN). Packet switching divides data into packets that share links, improving efficiency. Datagram switching is connectionless, while Virtual Circuits are connection-oriented.

Answer framework: Compare 'Circuit Switching' and 'Packet Switching' in a conceptual table; explain the 'Datagram' approach to routing.

Important point: Packet loss and jitter can affect real-time applications. Do not use packet-switched networks for time-critical control without authorised Quality of Service (QoS) mechanisms.

Explain Medium Access Control (MAC) and state one correct method, application or precaution. —

MAC protocols coordinate access to a shared medium. CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is used in wired Ethernet. CSMA/CA (Collision Avoidance) is used in wireless networks (Wi-Fi) to handle the 'Hidden Node' problem.

Answer framework: Describe the 'CSMA/CD' process conceptually; explain why 'CSMA/CA' is used in wireless LANs instead of collision detection.

Important point: Collisions reduce network efficiency. All shared-medium designs must follow authorised collision-domain limits and access-control rules.

Module 4 — Network and Transport Layers

Explain IP Addressing and Routing and state one correct method, application or precaution. —

The Network Layer handles end-to-end delivery. IP addresses identify devices. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses to handle the growing number of devices. Routing algorithms determine the best path for data.

Answer framework: Differentiate between 'IPv4' and 'IPv6' conceptually; explain the purpose of 'Subnetting' in IP networks.

Important point: Incorrect IP configuration can cause network isolation or conflicts. All IP planning must follow authorised address-allocation and subnetting schemes.

Explain Transport and Application Protocols and state one correct method, application or precaution. –

The Transport Layer provides process-to-process delivery. TCP is connection-oriented and reliable, while UDP is connectionless and fast. Congestion control prevents network collapse. Application layer protocols like HTTP (web) and DNS (naming) provide user services.

Answer framework: Compare 'TCP' and 'UDP' in a conceptual table; describe the role of 'DNS' in the internet.

Important point: Network congestion can lead to significant delays. Do not deploy high-traffic applications without authorised congestion-avoidance and load-balancing strategies.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Data Communication and Reference Models.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Physical and Data Link Layers.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Switching and Medium Access.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Network and Transport Layers.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.

2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Data communication components; Network topologies: Star, Mesh, Ring, Bus; Network categories: LAN, MAN, WAN; OSI Reference Model; TCP/IP Model; Comparison of models..
2. Develop a complete answer or practical plan for: Transmission media: Guided (Twisted pair, Coaxial, Fiber) and Unguided; Data Link Layer: Framing, Error detection (CRC, Parity), Flow control (Stop-and-wait, Sliding window); HDLC basics..
3. Develop a complete answer or practical plan for: Switching techniques: Circuit switching, Packet switching (Datagram and Virtual circuit); Medium Access Control (MAC): ALOHA, CSMA/CD, CSMA/CA; Ethernet standards; Wireless LAN (802.11)..
4. Develop a complete answer or practical plan for: Network Layer: IP Addressing (IPv4 vs IPv6), Subnetting; Routing algorithms: Distance Vector, Link State; Transport Layer: TCP and UDP features; Congestion control; Application layer overview (HTTP, DNS)..

LAST-MILE REVISION

Quick Revision

Module 1: Data Communication and Reference Models

Data communication components; Network topologies: Star, Mesh, Ring, Bus; Network categories: LAN, MAN, WAN; OSI Reference Model; TCP/IP Model; Comparison of models.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Physical and Data Link Layers

Transmission media: Guided (Twisted pair, Coaxial, Fiber) and Unguided; Data Link Layer: Framing, Error detection (CRC, Parity), Flow control (Stop-and-wait, Sliding window); HDLC basics.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Switching and Medium Access

Switching techniques: Circuit switching, Packet switching (Datagram and Virtual circuit); Medium Access Control (MAC): ALOHA, CSMA/CD, CSMA/CA; Ethernet standards; Wireless LAN (802.11).

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Network and Transport Layers

Network Layer: IP Addressing (IPv4 vs IPv6), Subnetting; Routing algorithms: Distance Vector, Link State; Transport Layer: TCP and UDP features; Congestion control; Application layer overview (HTTP, DNS).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Network Fundamentals	Data communication involves the exchange of data between two nodes.
Reference Models (OSI and TCP/IP)	Reference models provide a framework for network communication.
Transmission Media	Transmission media carry the signal from sender to receiver.
Data Link Control	The Data Link Layer ensures reliable point-to-point communication.
Switching Techniques	Switching directs data through a network.
Medium Access Control (MAC)	MAC protocols coordinate access to a shared medium.
IP Addressing and Routing	The Network Layer handles end-to-end delivery.
Transport and Application Protocols	The Transport Layer provides process-to-process delivery.

LEARNING RESOURCES

References

Recommended telecommunication networks references

1. Behrouz A. Forouzan, Data Communications and Networking.
2. Andrew S. Tanenbaum, Computer Networks.
3. William Stallings, Data and Computer Communications.
4. James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach.

Approved public telecommunication networks sources

- <https://nptel.ac.in/courses/106105183>
- <https://www.youtube.com/watch?v=IPvYjXCsTg8>

These sources provide the verified study basis while official Course 5043C detail is unavailable; they do not replace official carrier designs, authorised spectrum management, certified equipment specifications or authorised strategic communication plans.